

1 Question (20 points) (Java language)

Suppose you are given two sorted arrays A and B. Write a function that finds the number of elements in $A \triangle B = (A - B) \cup (B - A)$ (the elements that are in A but not in B and the elements that are in B but not in A) in $\mathcal{O}(N)$ time.

```
1 int symmetric(int[] A, int[] B)
```

2 Question (15 points) (C++ language)

Write a function that sorts an array of N elements with respect to their first digits. The time complexity of your function should be $\mathcal{O}(N)$.

1 `void sortFirstDigit (int* A, int N)`

3 Question (15 points) (Java language)

```
1 class Node {  
2     int data;  
3     Node next;  
4 }  
5 class LinkedList {  
6     Node head;  
7     Node tail;  
8 }
```

Write a function that will delete all **prime** nodes that is their data field is prime such as 2, 3, 5, 7, etc.

```
1 void deletePrimes()
```

4 Question (15 points) (C++ Language)

```
1 class Tree {  
2     TreeNode* root;  
3 }  
4 class TreeNode {  
5     TreeNode* left;  
6     TreeNode* right;  
7     int data;  
8 }
```

Write a class method in `TreeNode` class that returns the number of **leftist** nodes in a binary tree. A node is **leftist** if

- Its left child's data is larger than its right child's data or,
- It has only left child.

```
1 int leftist ()
```

5 Question (20 points) (Java language)

```
1 public class Edge {  
2     int from;  
3     int to;  
4     int weight;  
5     Edge next;  
6 }  
7 public class EdgeList {  
8     Edge head;  
9     Edge tail;  
10 }  
11 public class Graph {  
12     int vertexCount;  
13     EdgeList[] edges;  
14 }
```

A graph represents a ring topology if all the nodes create a circular path. Each node is connected to two others, like points on a circle. Write a class method in Graph class for adjacency list representation which checks if the corresponding graph is circular or not.

```
1 boolean isCircular ()
```

6 Question (15 points)

Given the following function

```
1 int algorithm(int N){  
2     int sum = 0;  
3     if (N == 0)  
4         return 0;  
5     if (N % 2 == 0){  
6         for (i = 0; i < N; i++)  
7             for (int j = 0; j < N; j++)  
8                 sum++;  
9     } else {  
10        for (int j = 0; j < N; j++)  
11            sum++;  
12    }  
13    return algorithm(N - 1) + sum;  
14 }
```

What is the time complexity of algorithm? Extract the recurrence equation for the time complexity. Solve the recurrence equation.